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10 JUNE 1952
SUPERSEDING
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10 MAY 1950

MILITARY SPECIFICATION

CHARGES, BURSTER, FOR SHELL, LOADING, ASSEMBLING, AND PACKING

This specification was approved by the Departments of the Army, the Navy, and the Air Force for the use of procurement services of the respective Departments.

1. SCOPE

1.1 Scope. — This specification covers the loading, assembling and packing of burster charges for shell.

1.2 Classification.—This specification covers all standard or authorized types and the following classes of burster charges:

Class A—Pressed tetryl charges.
Class B—Cast tetrytol charges.

2. APPLICABLE SPECIFICATIONS, STANDARDS, DRAWINGS, AND SPECIFICATIONS

2.1 The Ordnance Corps drawing pertaining to the burster charge under contract and all drawings and specifications referenced thereon, of the issue in effect on date of invitation for bids, form a part of this specification.

(Copies of specifications and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.—Material and parts shall conform to the applicable drawings and specifications.

3.2 The burster charge shall meet all requirements specified on the drawings and in applicable specifications.

3.3 Static functioning. — The burster charge shall detonate with high order when fired statically. A lot of burster charges will be considered to meet this requirement if samples selected and tested as prescribed in 4.7 meet the acceptance provisions prescribed therein.

3.4 Workmanship.—All parts shall be free from chips, dirt, grease, and other foreign material. The cleaning method used shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agents used.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Lot. — A lot shall consist of 20,000 burster charges, maximum. A smaller quantity may constitute a lot at the discretion of the inspector. Two thousand plus or minus 500 shall constitute the first lot manufactured on any contract, except when a new contract immediately follows the manufacture of satisfactory material of the same character. A lot shall consist of one type and one class of burster charge.

4.2 Sampling procedure.—All samples shall be selected by the inspector. Ten bursters shall be selected from each lot for tests as specified in the following paragraphs. Sampling of component materials is prescribed in paragraphs covering tests of the materials.

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4.3 **Inspection.**—Inspection shall be as specified in Specification MIL–G–2550.

4.4 **Determination of graphite and/or metallic stearate content of tetryl pellets.**—Ten pellets selected by the inspector from each lot before reconsolidation into the burster charge shall be tested as follows:

4.4.1 *Procedure.*—Remove the outer surface from the sample pellets to obtain a composite sample of about 5 grams (gms.) from the interior of the pellets. Transfer this weighed portion of approximately 5 gms. of the sample to a tared filtering crucible and extract with four 10-milliliter (ml.) portions of acetone, maintaining each portion in contact with the residue for approximately 5 minutes before applying suction, breaking up any lumps which may persist and washing down the sides of the crucible with each 10-ml. portion of acetone to insure complete removal of the tetryl. Aspirate the residue until it is almost free from acetone and dry at 100° to 105°C. for 1 hour. Calculate the weight of residue to percentage of graphite and/or metallic stearate in the sample.

4.4.2 *Retest.*—A retest, using double the number of samples used in the original test may be made if requested by the contractor provided the analysis indicates that the graphite and/or metallic stearate content exceeds the amount prescribed on the drawing by not more than 0.25 percent.

4.5 **Analysis of composition (Class B charge).**

4.5.1 *Trinitrotoluene.*—Transfer a weighed portion of 0.35 gm. of the dry well ground sample to a previously dried and weighed 100-ml. volumetric flask having a short neck and ground-glass stopper. Add approximately 100-ml. of carbon tetrachloride to the flask and heat the mixture to boiling in order to insure the solution of all TNT present. Stopper the flask, allow the flask and contents to cool to room temperature and weigh. Immerse the bulb of the flask in an ice bath for not less than 4 hours, agitating the contents every 15 minutes or half hour. At the end of this time take a sample of the supernatant liquid, which should be at a temperature between 0.0° and 0.5°C. in the following manner: Attach a 100-ml. pipette to a short piece of rubber tubing containing a cotton filter plug. Insert the end of the tubing into the supernatant liquid, quickly draw up as large a portion of the liquid as possible into the assembly and immediately transfer this to a previously dried and weighed Erlenmeyer flask having a tight-fitting stopper. Allow the flask and contents to warm up to room temperature, weigh, evaporate the carbon tetrachloride by means of a current of dry air and dry the residue at 60°C. for two hours and weigh. Calculate the TNT content of the sample as follows:

$$\text{Percentage of TNT} = \frac{Wa(100Wn - 0.0073Wc)}{Wc \times Wt}$$

where:

Wa = Total weight of carbon tetrachloride used.

Wc = Weight of carbon tetrachloride in portion of supernatant liquid taken.

Wn = Weight of residue from portion of supernatant liquid taken.

Wt = Weight of the tetrytol sample taken for analysis.

4.5.2 *Tetryl.*—Obtain the percentage of tetryl in the sample by subtracting the percentage of TNT from 100.

4.6 **Determination of specific gravity (Class B charge).**—One burster tube from each 100 to be loaded, shall be selected and marked "sample." This tube shall be weighed and the internal diameter and overall length carefully measured. The sample burster tube shall then be placed in the Loading Line and loaded under conditions identical in all respects with those employed in loading the lot. After loading, each sample tube shall be weighed and the specific gravity calculated as follows:

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$$\frac{Wt - We}{Ww} = \text{Specific Gravity}$$

We = Weight of empty tube in grams.

Wt = Weight of tetrytol loaded tube in grams.

Ww = Weight of water to fill tube in grams.

Ww is obtained by calculating the volume of the tube in cubic centimeters and multiplying this volume by 0.997. If the specific gravity complies with the drawing requirements for Class B burster tubes, the identification markings may be removed from the sample tubes, and these tubes included in the lot.

4.7 Static functioning test.—Ten burster charges selected by the inspector from each lot shall be subjected to test as follows:

4.7.1 *Procedure.*—Assemble each of the ten sample burster charges as shown on the figure (fig. 1, 2, 3, 4, 5, or 6) pertaining to the burster charge being tested. Embed the assembly in a sufficient quantity of sand in a container, so that on detonation there will be no fragments lost and no refragmentation caused by the fragments hitting the sides of the container. Fire the burster charge and recover all fragments. High order detonation shall be evidenced by a great number of small fragments of the burster casing and the burster tube, and by the absence of any unexploded tetryl, or tetrytol.

4.7.2 *Acceptance.*—The lot shall be considered acceptable provided the burster charges detonate with high order. If 9 of the 10 burster charges meet this requirement, the lot shall be considered eligible for retest under the provisions of 4.7.3. If less than 9 of the 10 burster charges meet this requirement, the lot shall be rejected.

4.7.3 *Retest.*—A retest as provided for in 4.7.2 may be made if requested by the contractor, using 20 burster charges. The lot shall be considered acceptable on retest provided the 20 burster charges detonate with high order.

5. PREPARATION FOR DELIVERY

5.1 Packing.—Packing shall be as specified on the applicable drawing.

5.2 Labeling and marking.—Labeling and marking shall be as specified on the applicable drawing.

5.3 Data cards.—Data cards, 5 by 8 inches, made of commercial manila tag board, approximately 200 pounds ream weight, shall be prepared for each lot of accepted charges to provide the information specified below. Quantities (approximately 50 cards per lot) and distribution of data cards shall be in accordance with instructions issued by the Office, Chief of Ordnance.

AMMUNITION DATA CARD

Kind:_____ Ammunition Lot No._____

Class_____ Quantity in Lot_____

Drg. No.*_____ Spec. No.*_____

Metals Parts Manufactured by_____ Lot No._____

Filler_____ Manufactured by_____ Lot No._____

Loaded and Assembled by_____ Date_____

Contract No._____ Order No._____

Packed_____ in a box.

Remarks:_____

Certified by_____, Inspector.
Print here name of inspector, district, arsenal, or loading plant, and sub-district, where applicable.

* Include revision and amendment.

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6. NOTES

6.1 Satisfactoriness of burster charge loading.—X-ray of approximately one of every 100 burster charges has been found to be dependable method of determining satisfactoriness of burster charge loading.

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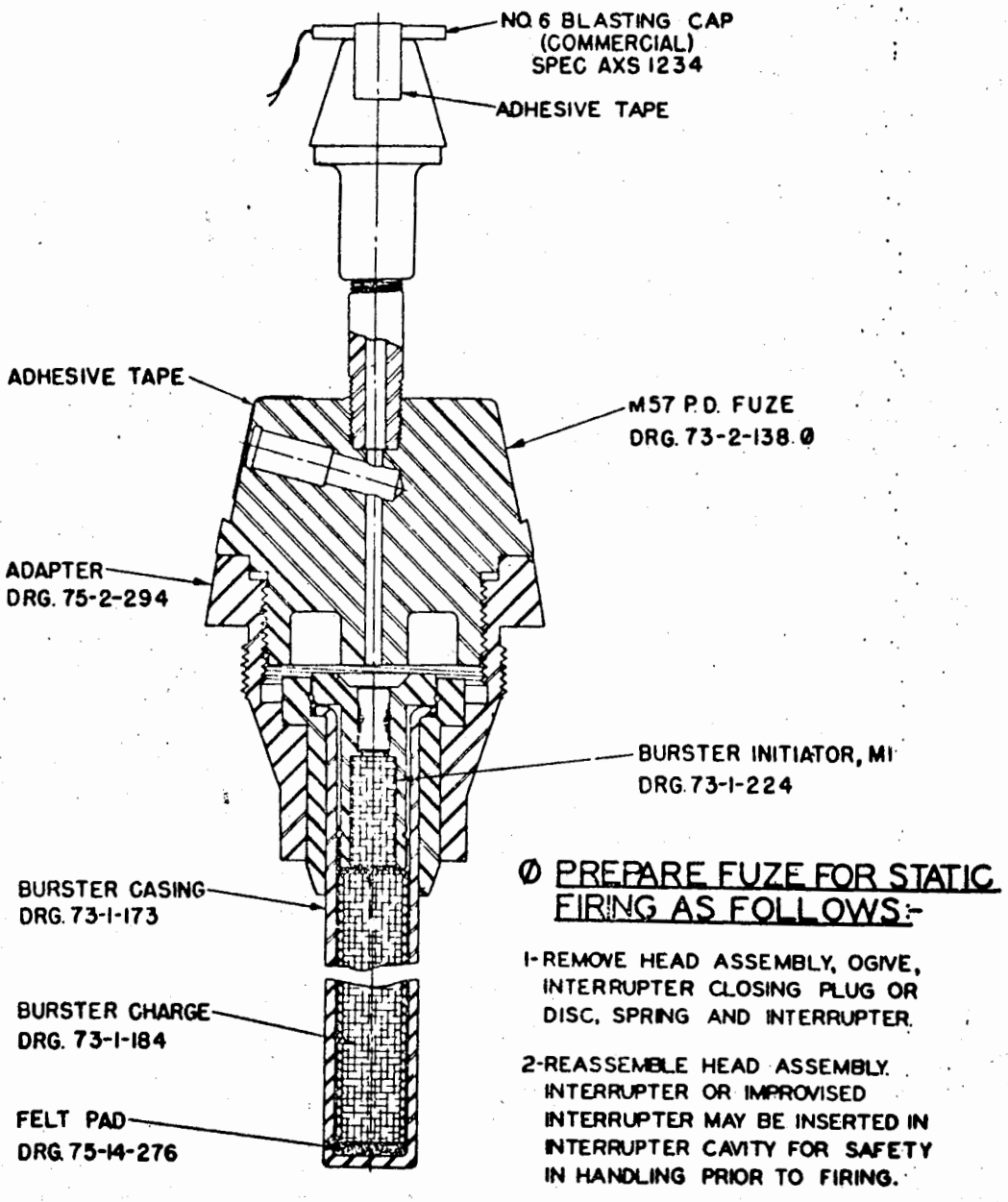
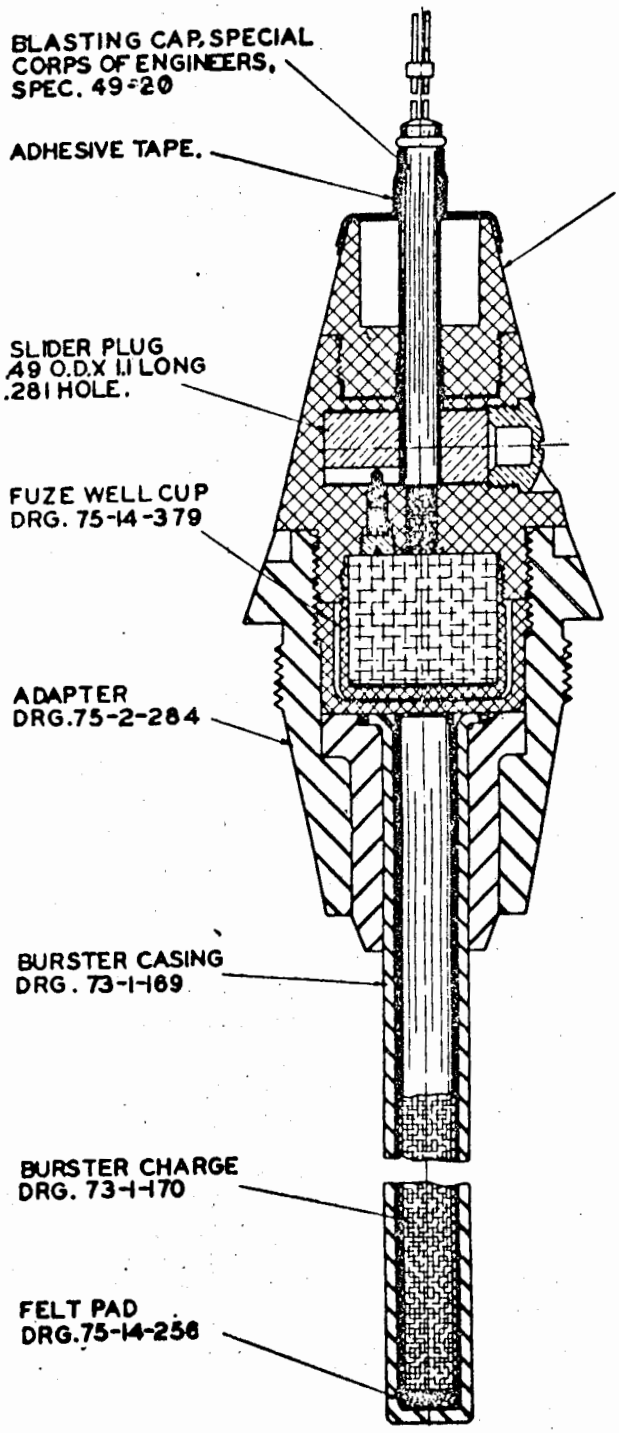


FIGURE 1.—*Assembly for static testing 75-mm burster charge.*

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ØPREPARE FUZE FOR STATIC FIRING AS FOLLOWS:-

1- REMOVE BOOSTER CUP AND PELLET.
2- REMOVE FIRING PIN ASSEMBLY AND FIRING PIN SPRING.
3- REMOVE SAFETY WIRE, SETBACK PLUG, SPRING AND PIN.
4- REMOVE SAFETY PIN AND SLIDER.
5- INSERT SLIDER PLUG.
6- ENLARGE HOLE THROUGH CENTER OF FUZE HEAD TO .281 DIA. TO ACCOMMODATE BLASTING CAP.
7- REASSEMBLE FUZE HEAD BOOSTER PELLET AND CUP.

FIGURE 2.—*Assembly for static testing 81-mm burster charge.*

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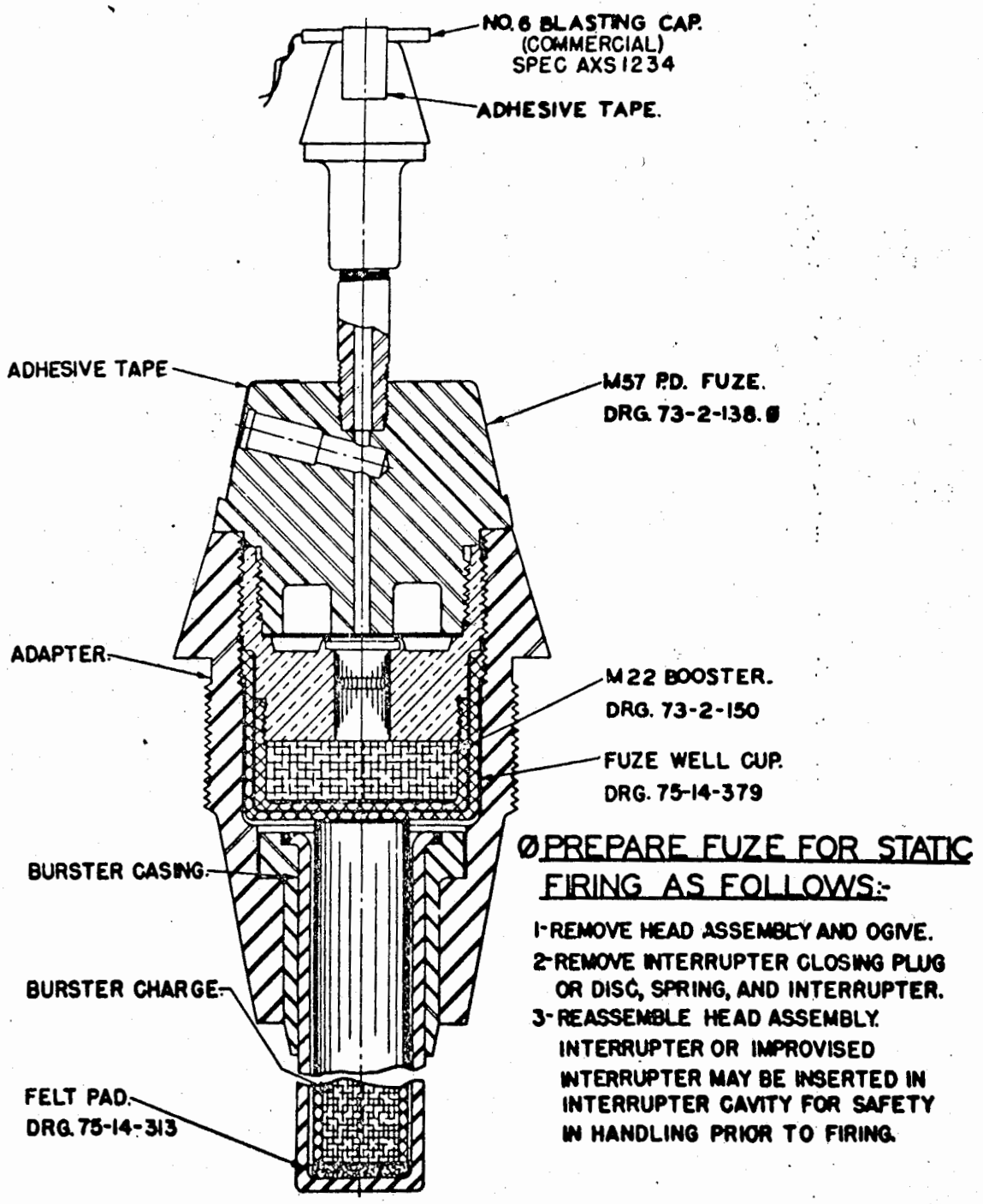


FIGURE 8.—*Assembly for static testing 105-mm, M60, 155-mm 8-inch and 240-mm burster charges.*

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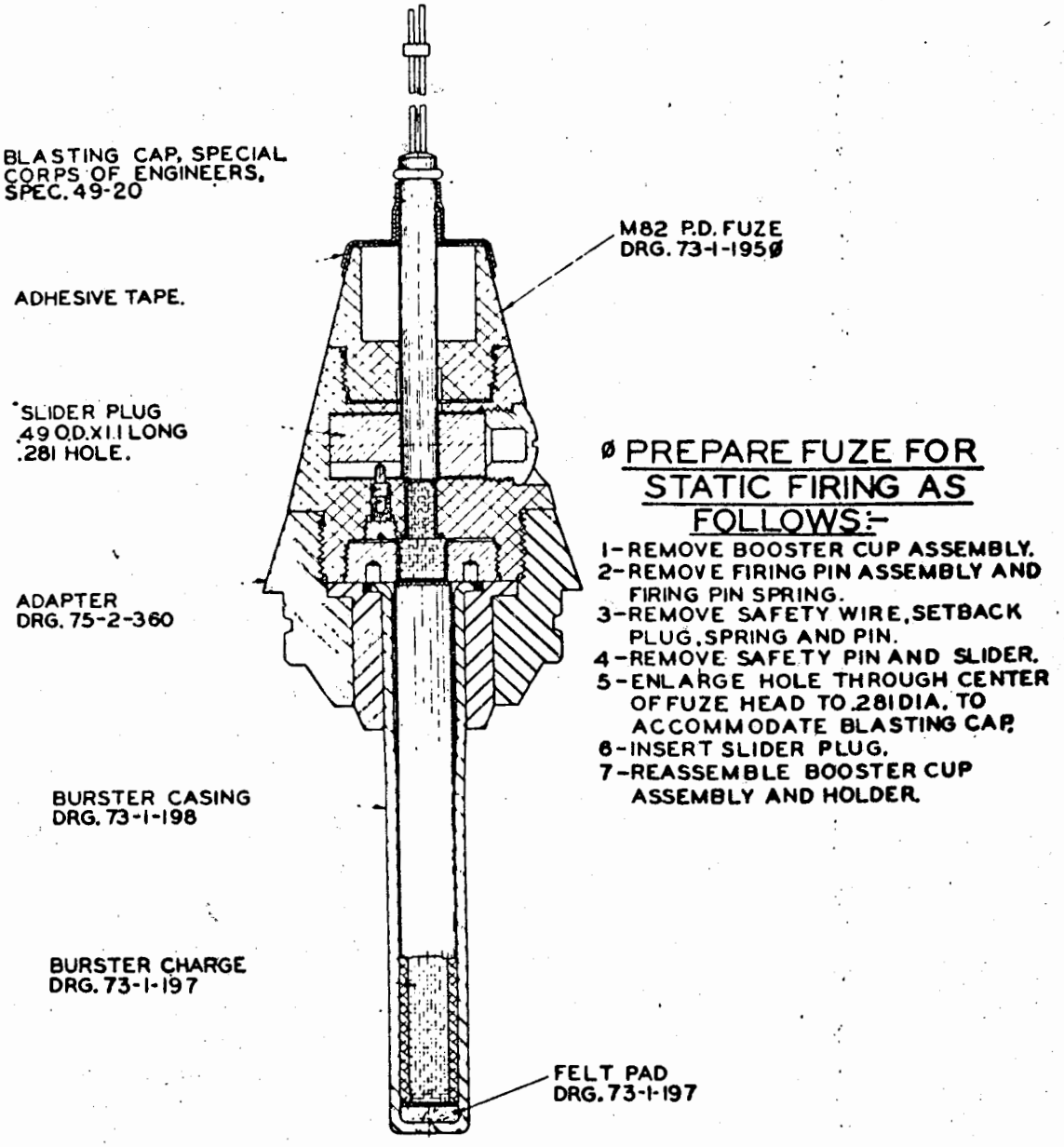


FIGURE 4.—*Assembly for static testing 60-mm smoke shell burster charge.*

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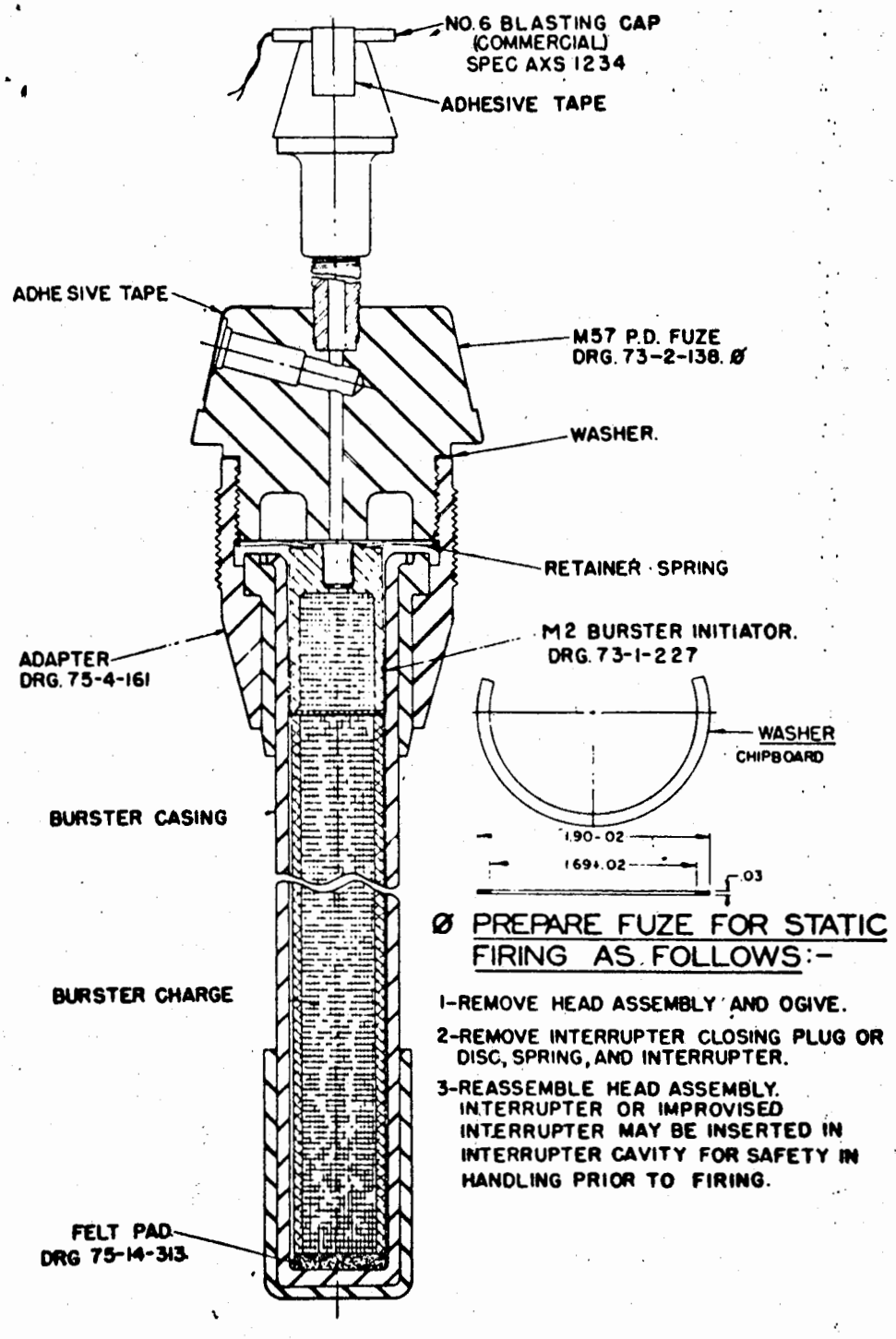
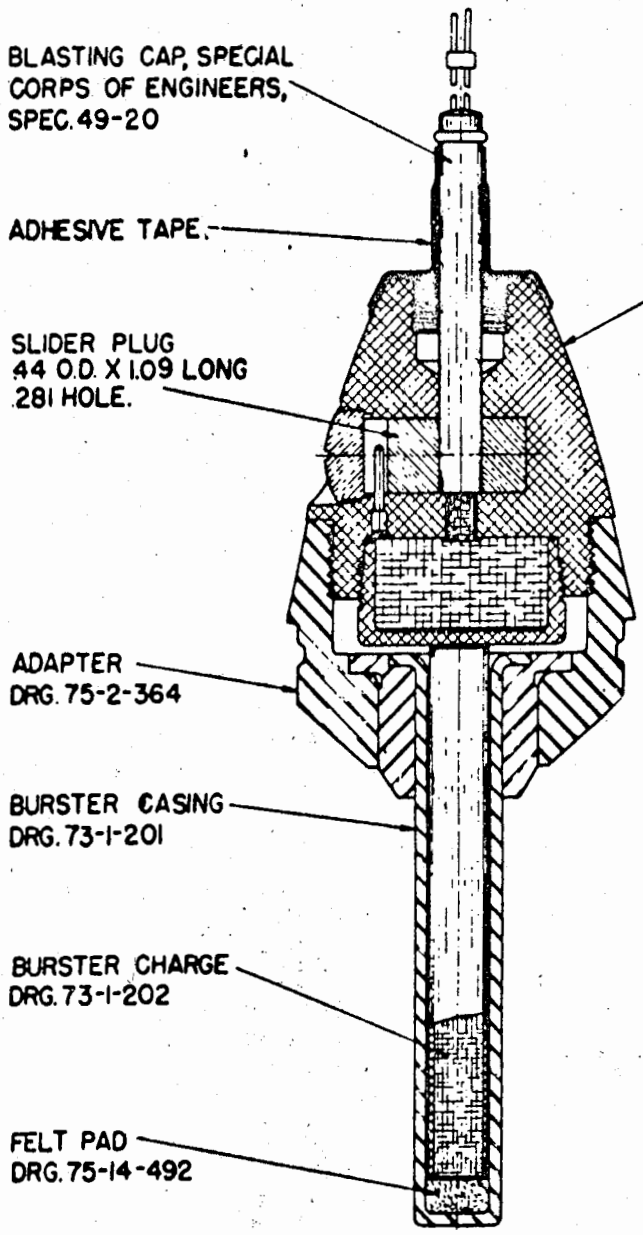


FIGURE 5.—*Assembly for static testing 76-mm, 90-mm, 105-mm T46 and 120-mm burster charges.*

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ØPREPARE FUZE FOR STATIC FIRING AS FOLLOWS:-

1- REMOVE BOOSTER CUP AND PELLET.
2- REMOVE HEAD ASSEMBLY.
3- REMOVE SLIDER PIN, ROTOR, ROTOR LOCK AND SPRING.
4- INSERT SLIDER PLUG AND REPLACE SLIDER PIN.
5- ENLARGE HOLE THROUGH CENTER OF FUZE BODY TO .281 DIA. TO ACCOMODATE BLASTING CAP.
6- REASSEMBLE BOOSTER PELLET AND CUP.

FIGURE 6.—*Assembly for static testing 57-mm burster charge.*

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